
INTERNAL BUILD INSTRUCTIONS

Procurement Intelligence Agent

AI Division – Saudi Construction & Infrastructure

1. PURPOSE (READ FIRST – NON-NEGOTIABLE)

We are building a **Procurement Intelligence Agent**, not a buyer and not an ERP replacement.

The agent's job is to:

- Prepare procurement actions
- Control procurement workflows
- Enforce Saudi compliance
- Provide full visibility and auditability
- Hand off **approved** data to Finance / ERP

AI prepares. Humans approve. Systems execute.

If the system:

- Commits spend
 - Skips approvals
 - Hides risk
- ☞ **It has failed.**
-

2. POSITION IN THE OVERALL SYSTEM

Upstream

- RFQ & Tender Intelligence Agent
(Approved scope, quantities, assumptions, drawings, BOQs)

Downstream

- Finance Agent
- ERP (SAP / Oracle / MS Dynamics via adapter)

Side systems

- CRM (visibility & approvals only)
 - Secure object storage (S3 / Azure Blob)
-

3. CORE PRINCIPLES (ENGINEERING GUARDRAILS)

1. ✗ No autonomous purchasing
 2. ✗ No negotiation
 3. ✗ No ERP logic duplication
 4. ✓ Full traceability to source documents
 5. ✓ Human approval at every commitment point
 6. ✓ Saudi compliance enforced before progression
-

4. WHAT INPUTS THE AGENT RECEIVES (STRICT)

The Procurement Agent ONLY accepts:

- Approved Tender ID
- Approved scope packages (by trade)
- Validated quantities / BOQs
- Assumptions & exclusions
- Approved supplier list (AVL)
- Budget & cost codes
- Approval matrix

✗ If approval status $\neq$ APPROVED $\rightarrow$ STOP and request approval

✗ If data is missing $\rightarrow$ Ask ONE question only

5. PROCUREMENT WORKFLOW (MANDATORY FLOW)

State Machine (Must Be Implemented)

READY $\rightarrow$ RFQ_PREPARED $\rightarrow$ RFQ_SENT $\rightarrow$ QUOTES_RECEIVED
 $\rightarrow$ NORMALIZED $\rightarrow$ COMPLIANCE_CHECKED $\rightarrow$ DECISION_PACK_READY
 $\rightarrow$ APPROVED $\rightarrow$ HANDED_TO_FINANCE

Rework loops:

REJECTED $\rightarrow$ REQUOTE_REQUESTED $\rightarrow$ RFQ_SENT

6. SUPPLIER MANAGEMENT RULES

Supplier Mapping

- Use Approved Vendor List (AVL) only
- Filter suppliers by:
 - Trade
 - Geography
 - Certifications
 - Project rules

✗ Never auto-add new suppliers

✗ Never issue RFQs to inactive vendors

7. RFQ PACK BUILDER (CRITICAL)

What to Build

For each supplier:

- Create a **vendor-scoped RFQ pack**
- Include only:
 - Relevant drawings
 - Relevant specs
 - Relevant BOQ lines
 - Standards applicable to that scope
- Generate a **manifest** listing all included files

Hard Rule

⊘ **Zero information leakage between suppliers**

This must be tested.

8. RFQ ISSUANCE & TRACKING

The agent must:

- Issue RFQs (email / portal adapter)
- Track:
 - Sent
 - Opened
 - Responded
 - Overdue

- Send reminders
- Escalate overdue RFQs to humans

✗ AI must not discuss pricing or negotiate

9. QUOTE INGESTION & EXTRACTION

For every supplier quote:

- Store original file as read-only
- Extract:
 - Line items
 - Quantities
 - Unit rates
 - Lead time
 - Exclusions
 - Payment terms
 - VAT treatment

Each extracted field must include:

- Confidence score
 - Source reference (doc + page)
-

10. QUOTE NORMALISATION (APPLE-TO-APPLE)

All quotes must be normalized into a **canonical structure**:

```
line_id | description | uom | qty | unit_rate | total
status: MATCHED / PARTIAL / MISSING / EXTRA
```

The agent must:

- Align quantities
- Flag scope gaps
- Flag exclusions
- Highlight deviations

✗ Never “fix” quantities automatically

11. SAUDI PROCUREMENT COMPLIANCE (MANDATORY)

Must Enforce:

- VAT / ZATCA registration
- Correct VAT rate application
- Supplier certifications
- Local content rules (if applicable)
- Project-specific rules:
 - Aramco
 - SEC
 - RCJY
 - NEOM / Red Sea / Diriyah

Compliance Output

- Rules passed
- Rules failed
- Severity (HIGH blocks / MEDIUM review)

✗ HIGH severity = cannot proceed

12. FRAUD & RISK CONTROLS (DO NOT SKIP)

Bank Detail Change Detection

- If supplier bank details differ from vendor master:
 - HARD BLOCK
 - Require human verification

Procurement Risk Score

Compute per supplier:

- Compliance risk
- Delivery risk
- Scope risk
- Commercial risk

Output:

Supplier A: LOW
Supplier B: MEDIUM
Supplier C: HIGH

13. APPROVAL ORCHESTRATION

Approval Layers (Configurable)

- Engineering confirmation
- Procurement approval
- Commercial approval
- Project Director approval

The agent must:

- Route approvals
- Track delays
- Escalate bottlenecks

✗ No approval → No finance handover

14. HANDOVER TO FINANCE (STRICT CONTRACT)

Only after approval:

- Generate **PO Readiness Payload**
- Include:
 - Selected supplier
 - Approved pricing
 - Terms
 - Compliance status
 - Audit references

Finance handles:

- PO creation
- 3-way matching
- Payments

15. CRM INVOLVEMENT (READ-ONLY)

CRM is used for:

- Project visibility
- Approval tracking
- Stakeholder comments

CRM must NOT:

- Store pricing logic
- Issue RFQs
- Trigger ERP actions

16. STORAGE & SECURITY

- Store all documents in secure object storage
- Originals are read-only
- All actions logged with:
 - Who
 - When
 - What
 - Why
 - Source reference

Saudi data residency rules must be respected where required.

17. HUMAN REVIEW UI (REQUIRED FOR MVP)

Must include:

- Quote comparison table
 - Compliance flags
 - Risk scores
 - Approval buttons
 - Audit trail view
 - Export decision pack (PDF / Excel)
-

18. WHAT IS EXPLICITLY OUT OF SCOPE

- ⊗ Supplier negotiation
 - ⊗ Contract signing
 - ⊗ Autonomous PO issuance
 - ⊗ ERP replacement
 - ⊗ Pricing strategy decisions
-

19. ACCEPTANCE CRITERIA (HOW WE TEST)

The system is **NOT DONE** unless:

- RFQ packs are vendor-scoped and leak-proof
- Quotes are normalized and traceable
- VAT non-compliance blocks progress

- No finance handover without approvals
 - Audit trail shows every step
 - Re-quote and addendum flows work
-

20. SUCCESS METRICS

- Reduced procurement cycle time
 - Fewer scope disputes
 - Strong audit outcomes
 - Faster PO readiness
 - High user trust and adoption
-